A Conditional Mutual Information Framework for Interaction-Order Contributions in Learned Representations

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Draft extension of the cross-fitted CMI framework

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1 Introduction

This document extends the cross-fitted conditional mutual information (CMI) framework from *scores* to *interaction-order contributions* in learned representations.

The central question is:

> How much unique predictive information about a target Y is contributed by the order-k interaction features, beyond what is already explained by the baseline representation and all lower-order interactions?

This allows us to move from score comparison to a principled information-theoretic analysis of *representation complexity*. In particular, it provides a formal way to evaluate whether pairwise interactions are sufficient, whether higher-order interactions add predictive value, and which interaction orders matter most.

2 Setup and Notation

Let

- $Y \in \{0, 1\}$ be a binary target,
- R denote a baseline representation,
- $\Phi_1, \Phi_2, \ldots, \Phi_K$ denote feature blocks corresponding to interaction orders $1, 2, \ldots, K$,
- $\Phi_{<k} := (\Phi_1, \ldots, \Phi_{k-1})$,
- $\Phi_{\leq k} := (\Phi_1, \ldots, \Phi_k)$.

Interpretation:

- Φ_1 may contain unary / linear features,
- Φ_2 may contain pairwise interactions,
- Φ_3 may contain triplet interactions,
- and so on.

The role of the baseline representation R is to prevent higher-order blocks from receiving credit for predictive information that is already available in the underlying representation.

2.1 The Role and Choice of the Baseline Representation R

The choice of R is one of the most consequential design decisions in the framework, because it determines the precise meaning of the word "unique" in Δ_k. Concretely, $\Delta_k = I(Y; \Phi_k \mid R, \Phi_{<k})$ answers the question: *how much does Φ_k add, given that R and all lower-order blocks are already known?* Different choices of R give genuinely different answers, each with a distinct scientific interpretation.

$R = \varnothing$ (empty baseline). Δ_k measures the incremental value of Φ_k beyond the lower-order feature blocks alone. This is the cleanest setting for synthetic validation, where no learned representation is involved.

$R =$ a frozen backbone embedding. Δ_k measures how much Φ_k adds beyond what the backbone already captures. This is the appropriate choice when evaluating whether explicit interaction features provide complementary signal on top of a pretrained model. Importantly, if the backbone was *trained using* Φ_k as part of its input or objective, then R and Φ_k are statistically entangled: conditioning on R partially conditions on Φ_k itself, leading to systematic underestimation of Δ_k. To avoid this circularity, R should ideally come from a backbone trained *without explicit knowledge* of the Φ_k blocks being evaluated.

$R =$ the output of a specific intermediate layer ℓ. This choice enables an *interaction emergence profile*: by sweeping ℓ across depth, one can track $\Delta_k(\ell) = I(Y; \Phi_k \mid \text{layer}_\ell, \Phi_{<k})$ and identify at which depth the network ceases to benefit from the explicit interaction block. At shallow layers, $\Delta_k(\ell)$ is expected to be large (the network has not yet encoded order-k structure); at deeper layers it should decay as the network learns to represent interactions internally.

Remark 1 (Effect of richer R on estimator requirements)**.** *As R becomes higher-dimensional or more informative, the auxiliary model for $P(Y \mid R, \Phi_{<k})$ must be flexible enough to use R effectively. A misspecified or underfit auxiliary model will underestimate $H(Y \mid R, \Phi_{<k})$, inflating Δ_k. In practice, richer R demands more expressive auxiliary estimators (e.g. gradient-boosted trees rather than logistic regression), or careful dimensionality reduction of R before estimation.*

3 Theoretical Foundation

3.1 Order-k Conditional Mutual Information

Definition 1 (Order-k Incremental Predictive Information)**.** *The incremental predictive information carried by the order-k interaction block is defined as*

$$\Delta_k := I(Y; \Phi_k \mid R, \Phi_{<k}). \tag{1}$$

This quantity measures the reduction in uncertainty about Y when Φ_k is observed, given that the baseline representation R and all lower-order interaction blocks $\Phi_{<k}$ are already known.

Remark 2 (Relativity of Δ_k to the choice of R)**.** *Δ_k is not an absolute property of Φ_k; it is a property of Φ_k* relative to R. *If R already contains the information in Φ_k (e.g. because R was produced by a model trained with Φ_k as input), then $\Delta_k \approx 0$ not because Φ_k is uninformative about Y, but because R has already absorbed that information. Conversely, if R is an early-layer or task-agnostic embedding, Δ_k may be large even for interaction orders that a deeper backbone would have implicitly learned. The same Φ_k block can yield very different Δ_k values depending on which R is used, and comparing Δ_k across experiments is only meaningful when R is held fixed.*

3.2 Entropy Reduction Identity

Proposition 1 (Entropy Reduction Identity). *For each $k \geq 1$,*

$$\Delta_k = H(Y \mid R, \Phi_{<k}) - H(Y \mid R, \Phi_{\leq k}). \tag{2}$$

Proof. By the definition of conditional mutual information,

$$I(Y; \Phi_k \mid R, \Phi_{<k}) = H(Y \mid R, \Phi_{<k}) - H(Y \mid R, \Phi_k, R, \Phi_{<k}).$$

Since $(\Phi_k, \Phi_{<k}) = \Phi_{\leq k}$, we obtain

$$I(Y; \Phi_k \mid R, \Phi_{<k}) = H(Y \mid R, \Phi_{<k}) - H(Y \mid R, \Phi_{\leq k}).$$

□

3.3 Interpretation

Equation (2) shows that Δ_k is the *incremental conditional entropy reduction* obtained by adding the order-k features to a predictor that already has access to the baseline representation and all lower-order interactions.

Thus:

- $\Delta_k > 0$ means order k contributes unique predictive information,
- $\Delta_k \approx 0$ means order k is conditionally redundant given lower orders,
- larger Δ_k means higher unique relevance of order k for predicting Y.

3.4 Binary Entropy Formulation

For binary targets, define the binary entropy function

$$H_b(p) = -p \log p - (1-p)\log(1-p), \qquad p \in (0,1). \tag{3}$$

Theorem 1 (Order-k CMI via Binary Entropy Reduction). *For binary Y,*

$$\Delta_k = \mathbb{E}[H_b(P(Y=1 \mid R, \Phi_{<k})) - H_b(P(Y=1 \mid R, \Phi_{\leq k}))]. \tag{4}$$

Proof. Apply Proposition (2) and use the fact that for binary Y,

$$H(Y \mid X) = \mathbb{E}\left[H_b(P(Y=1 \mid X))\right].$$

Substituting $X = (R, \Phi_{<k})$ and $X = (R, \Phi_{\leq k})$ yields the result. □

4 Decomposition Across Interaction Orders

Theorem 2 (Chain Rule Decomposition Across Orders). *For any $K \geq 1$,*

$$I(Y; \Phi_{\leq K} \mid R) = \sum_{k=1}^{K} I(Y; \Phi_k \mid R, \Phi_{<k}) = \sum_{k=1}^{K} \Delta_k. \tag{5}$$

Proof. Apply the chain rule of conditional mutual information:

$$I(Y; \Phi_1, \ldots, \Phi_K \mid R) = \sum_{k=1}^{K} I(Y; \Phi_k \mid R, \Phi_1, \ldots, \Phi_{k-1}).$$

By definition, each summand is Δ_k. □

Equation (5) motivates the *interaction-order profile*

$$(\Delta_1, \Delta_2, \ldots, \Delta_K),$$

which quantifies how predictive information is distributed across interaction orders.

5 Estimation via Cross-Entropy Reduction

5.1 Probabilistic Model Approach

To estimate Δ_k, we fit two probabilistic models:

$$\hat{p}_{<k}(r, \phi_{<k}) \approx P(Y = 1 \mid R = r, \Phi_{<k} = \phi_{<k}), \tag{6}$$
$$\hat{p}_{\leq k}(r, \phi_{\leq k}) \approx P(Y = 1 \mid R = r, \Phi_{\leq k} = \phi_{\leq k}). \tag{7}$$

The first is the *restricted model*, which excludes order k. The second is the *full model*, which includes order k.

5.2 Empirical Estimator

Suppose we observe i.i.d. data

$$\{(Y_i, R_i, \Phi_{1,i}, \ldots, \Phi_{K,i})\}_{i=1}^{n}.$$

Definition 2 (Empirical Order-k CMI Estimator)**.** *The empirical estimator of Δ_k is*

$$\hat{\Delta}_k = \frac{1}{n} \sum_{i=1}^{n} \left[H_b(\hat{p}_{<k}(R_i, \Phi_{<k,i})) - H_b(\hat{p}_{\leq k}(R_i, \Phi_{\leq k,i})) \right]. \tag{8}$$

5.3 Log-Loss Equivalence

Define the binary cross-entropy loss

$$\ell(y, p) = -y \log p - (1 - y) \log(1 - p). \tag{9}$$

Theorem 3 (Log-Loss Gain Equivalence)**.** *The empirical estimator in (8) is equivalent to the average held-out log-loss gain:*

$$\hat{\Delta}_k = \frac{1}{n} \sum_{i=1}^{n} \left[\ell(Y_i, \hat{p}_{<k}(R_i, \Phi_{<k,i})) - \ell(Y_i, \hat{p}_{\leq k}(R_i, \Phi_{\leq k,i})) \right]. \tag{10}$$

Proof. For binary targets, expected cross-entropy equals binary entropy when evaluated at the true conditional probability. Thus, the entropy reduction formulation in (8) is equivalent to reduction in expected log-loss, and empirically this is estimated by the sample average in (10). □

Equation (10) is the practical estimator we will use:

$$\text{incremental CMI of order } k \approx \text{held-out log-loss of restricted model} - \text{held-out log-loss of full model.}$$

6 Cross-Fitting for Unbiased Estimation

6.1 Why Cross-Fitting is Necessary

If the same data are used both to train $\hat{p}_{<k}, \hat{p}_{\leq k}$ and to evaluate (8), the resulting estimate is optimistically biased due to overfitting. This is especially problematic in high-dimensional representation spaces.

6.2 Cross-Fitted Estimator

Algorithm 1 Cross-Fitted Estimation of $\Delta_k = I(Y; \Phi_k \mid R, \Phi_{<k})$

Require: Dataset $\mathcal{D} = \{(Y_i, R_i, \Phi_{1,i}, \ldots, \Phi_{K,i})\}_{i=1}^{n}$, number of folds M, order k

Partition indices $\{1, \ldots, n\}$ into disjoint folds $F_1, \ldots, F_M$

for $m = 1$ to M **do**

  $\mathcal{D}_{\text{train}}^{(m)} \leftarrow \{i : i \notin F_m\}$

  $\mathcal{D}_{\text{test}}^{(m)} \leftarrow \{i : i \in F_m\}$

  Train restricted model $\widehat{P}_{<k}^{(m)}$ on inputs $(R, \Phi_{<k})$

  Train full model $\widehat{P}_{\leq k}^{(m)}$ on inputs $(R, \Phi_{\leq k})$

  for $i \in F_m$ **do**

    $\hat{p}_{<k,i}^{(m)} \leftarrow \widehat{P}_{<k}^{(m)}(R_i, \Phi_{<k,i})$

    $\hat{p}_{\leq k,i}^{(m)} \leftarrow \widehat{P}_{\leq k}^{(m)}(R_i, \Phi_{\leq k,i})$

  end for

$$\widehat{\Delta}_k^{(m)} \leftarrow \frac{1}{|F_m|} \sum_{i \in F_m} \left[\ell(Y_i, \hat{p}_{<k,i}^{(m)}) - \ell(Y_i, \hat{p}_{\leq k,i}^{(m)}) \right]$$

end for

$$\widehat{\Delta}_k^{\text{CF}} = \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta}_k^{(m)}$$

return $\widehat{\Delta}_k^{\text{CF}}$

Remark 3 (Circularity risk when R is a trained embedding)**.** *The cross-fitting procedure eliminates overfitting bias in the auxiliary models $\hat{p}_{<k}$ and $\hat{p}_{\leq k}$. However, it does* not *resolve the deeper issue that arises when R itself was trained on data that includes Φ_k: in that case, R already encodes part of the signal in Φ_k, so the restricted model (which sees R) effectively has partial access to Φ_k, and the estimated Δ_k will be biased downward regardless of the cross-fitting design. The remedy is to ensure that the backbone producing R was trained on a held-out task or without the explicit interaction features being evaluated. In the synthetic validation experiments (Section 13), this issue does not arise because $R = \varnothing$.*

6.3 Asymptotic Interpretation

Proposition 2 (Asymptotic Interpretation)**.** *Under regularity conditions and sufficient calibration of the auxiliary predictors,*

$$\widehat{\Delta}_k^{\text{CF}} \xrightarrow[n \to \infty]{} I(Y; \Phi_k \mid R, \Phi_{<k}). \tag{11}$$

Remark 4. *As in the original score-based framework, finite-sample bias depends on:*

- *the quality of the auxiliary conditional probability estimators,*
- *model calibration,*
- *sample size relative to representation dimension,*
- *and the cross-fitting design.*

Thus, $\widehat{\Delta}_k^{\mathrm{CF}}$ should be interpreted as a predictive plug-in estimate of incremental conditional information.

7 Order Selection and Adaptive Interaction Modeling

7.1 Order Relevance Profile

The vector

$$\boldsymbol{\Delta} = (\Delta_1, \Delta_2, \ldots, \Delta_K) \tag{12}$$

defines the *order relevance profile* of the task.

This profile answers:

- Which interaction order first contributes substantial predictive information?
- Are pairwise interactions sufficient?
- Do triplet or higher-order terms add unique signal beyond lower orders?

Remark 5 (Order profile is relative to R)**.** *The order relevance profile $\boldsymbol{\Delta}$ characterizes task complexity* relative to the baseline R. *A profile with large Δ_1 and small Δ_k for $k \geq 2$ may reflect either that the task is genuinely linear, or that R has already captured all pairwise and higher-order structure. When comparing profiles across architectures or tasks, R must be held fixed to ensure the comparisons are meaningful.*

7.2 Learning Which Orders Matter

Suppose we build a model with order-specific branches:

$$\Psi(x) = \sum_{k=1}^{K} \alpha_k(x)\, \Phi_k(x),$$

where $\alpha_k(x)$ are order gates.

The information-theoretic goal is to retain orders with high incremental value:

$$\Delta_k = I(Y; \Phi_k \mid R, \Phi_{<k}).$$

This suggests a principled order-selection rule:

- keep order k if Δ_k is large relative to its computational cost,
- prune order k if $\Delta_k \approx 0$ given lower orders.

Thus, the CMI framework provides a formal target for *adaptive interaction-order selection.*

8 Specialization to Clifford-Style Order-2 Interactions

For order 2, let Φ_2 denote the pairwise interaction block. In a Clifford-style architecture, Φ_2 may include both:

- symmetric / coherence terms (inner-product-like),
- antisymmetric / structural terms (wedge-like).

Then

$$\Delta_2 = I(Y; \Phi_2 \mid R, \Phi_1)$$

measures the unique predictive information carried by pairwise relational structure beyond the baseline representation and unary terms.

More finely, if $\Phi_2 = (\Phi_2^{\mathrm{sym}}, \Phi_2^{\mathrm{asym}})$, one may define

$$\Delta_2^{\mathrm{sym}} = I(Y; \Phi_2^{\mathrm{sym}} \mid R, \Phi_1), \tag{13}$$

$$\Delta_2^{\mathrm{asym}} = I(Y; \Phi_2^{\mathrm{asym}} \mid R, \Phi_1, \Phi_2^{\mathrm{sym}}), \tag{14}$$

to separate the predictive contribution of symmetric and antisymmetric pairwise structure.

Remark 6 (Choice of R for Clifford-style analysis)**.** *In the order-2 case, the choice of R has a direct impact on whether Δ_2 captures genuinely* relational *structure or merely richer unary representations. If R is a strong unary encoder (e.g. the output of a deep MLP applied independently to each input), then $\Delta_2 > 0$ provides clean evidence that pairwise structure adds information beyond what per-element processing can capture. If R is weak or absent, Δ_2 may absorb both unary and relational signal, making interpretation less precise. For the branch-wise decomposition into symmetric and antisymmetric terms, the ordering matters: Δ_2^{sym} is computed first, and Δ_2^{asym} measures what remains after the symmetric component is already known. Swapping the order would yield different values and a different attribution.*

9 Numerical Stability and Uncertainty Quantification

9.1 Clipped Binary Entropy

For numerical stability, define the clipped binary entropy

$$H_b^{\varepsilon}(p) = H_b(\mathrm{clip}(p, \varepsilon, 1-\varepsilon)), \qquad \varepsilon > 0. \tag{15}$$

9.2 Fold-Based Standard Error

Given fold-specific estimates $\widehat{\Delta}_k^{(1)}, \ldots, \widehat{\Delta}_k^{(M)}$, define

$$\mathrm{SE}\Big(\widehat{\Delta}_k^{\mathrm{CF}}\Big) = \sqrt{\frac{1}{M-1} \sum_{m=1}^{M} \Big(\widehat{\Delta}_k^{(m)} - \overline{\Delta}_k\Big)^2}, \qquad \overline{\Delta}_k = \frac{1}{M} \sum_{m=1}^{M} \widehat{\Delta}_k^{(m)}. \tag{16}$$

10 Interpretation

- $\Delta_k > 0$: order k provides unique predictive information beyond lower orders,
- $\Delta_k \approx 0$: order k is conditionally redundant,
- larger Δ_k: stronger evidence that the target depends on order-k interactions.

Because the estimator is based on cross-entropy reduction, all quantities are measured in nats. To convert to bits, divide by $\log 2$.

11 Conclusion

This framework generalizes cross-fitted conditional mutual information estimation from model scores to interaction-order contributions in learned representations.

The core quantity,

$$\Delta_k = I(Y; \Phi_k \mid R, \Phi_{<k}),$$

measures the unique predictive information carried by order-k interactions beyond the baseline representation and all lower-order structure.

Its cross-fitted log-loss estimator provides a practical and principled way to:

- quantify which interaction orders matter,
- compare pairwise vs higher-order interaction blocks,
- and guide adaptive interaction-order selection in neural architectures.

12 Comparison Framework

This section formalizes how the order-wise CMI quantities can be used to compare interaction orders, architectural branches, and alternative representation families.

12.1 Comparison of Interaction Orders

For two interaction orders $j < k$, the quantities

$$\Delta_j = I(Y; \Phi_j \mid R, \Phi_{<j}), \qquad \Delta_k = I(Y; \Phi_k \mid R, \Phi_{<k})$$

measure different conditional contributions, since the conditioning sets differ.

Thus, Δ_j and Δ_k should not be interpreted as direct substitutes, but rather as elements of an *incremental hierarchy*:

$$I(Y; \Phi_{\leq K} \mid R) = \sum_{m=1}^{K} \Delta_m.$$

The natural comparison questions are:

- At which order does the first large gain in predictive information occur?
- Does the cumulative gain saturate at low order?
- Is the marginal contribution Δ_k negligible for sufficiently large k?

Definition 3 (Cumulative Interaction Information). *For $k \geq 1$, define the cumulative interaction information up to order k as*

$$\Gamma_k := I(Y; \Phi_{\leq k} \mid R) = \sum_{m=1}^{k} \Delta_m. \tag{17}$$

The sequence $(\Gamma_1, \Gamma_2, \ldots, \Gamma_K)$ describes how predictive information accumulates as interaction complexity grows.

12.2 Comparison of Branches Within an Order

Suppose order k decomposes into branches,

$$\Phi_k = (\Phi_k^{(1)}, \ldots, \Phi_k^{(B)}).$$

For example, in a Clifford-style order-2 block one may separate:

$$\Phi_2 = (\Phi_2^{\mathrm{sym}}, \Phi_2^{\mathrm{asym}}),$$

corresponding to symmetric/coherence and antisymmetric/structural terms.

We define branch-wise incremental contributions sequentially:

$$\Delta_k^{(1)} := I(Y; \Phi_k^{(1)} \mid R, \Phi_{<k}), \tag{18}$$

$$\Delta_k^{(2)} := I(Y; \Phi_k^{(2)} \mid R, \Phi_{<k}, \Phi_k^{(1)}), \tag{19}$$

$$\vdots$$

$$\Delta_k^{(B)} := I(Y; \Phi_k^{(B)} \mid R, \Phi_{<k}, \Phi_k^{(1)}, \ldots, \Phi_k^{(B-1)}). \tag{20}$$

By the chain rule,

$$I(Y; \Phi_k \mid R, \Phi_{<k}) = \sum_{b=1}^{B} \Delta_k^{(b)}. \tag{21}$$

This provides a principled decomposition of the total order-k contribution into branch-specific conditional gains.

12.3 Comparison of Architectures

Consider two representation families, $\mathcal{A}$ and $\mathcal{B}$, producing interaction blocks

$$\Phi_1^{\mathcal{A}}, \ldots, \Phi_K^{\mathcal{A}}, \qquad \Phi_1^{\mathcal{B}}, \ldots, \Phi_K^{\mathcal{B}}.$$

Examples include:

- a Clifford-style order-2 architecture,
- a generic bilinear / quadratic interaction architecture,
- a tensorized higher-order interaction model,
- a standard FFN-based baseline.

For each architecture, define the order profile

$$\boldsymbol{\Delta}^{\mathcal{A}} = (\Delta_1^{\mathcal{A}}, \dots, \Delta_K^{\mathcal{A}}), \qquad \boldsymbol{\Delta}^{\mathcal{B}} = (\Delta_1^{\mathcal{B}}, \dots, \Delta_K^{\mathcal{B}}).$$

Then architecture comparison can be performed along at least three axes:

1. **Total information captured:**
$$\Gamma_K^{\mathcal{A}} \quad \text{vs.} \quad \Gamma_K^{\mathcal{B}}.$$
2. **Order efficiency:** compare Δ_k per parameter or per FLOP.
3. **Information concentration:** determine whether information is concentrated at low order or only appears at high order.

Remark 7 (Comparability requires a shared R)**.** *Architecture comparison via order profiles is only meaningful when both architectures are evaluated against the* same *baseline R. If $R^{\mathcal{A}} \neq R^{\mathcal{B}}$, differences in $\boldsymbol{\Delta}$ may reflect differences in the baselines rather than in the interaction blocks themselves. A natural choice is to use a shared, architecture-agnostic R (e.g. raw input features, or a frozen backbone trained independently of both architectures).*

Definition 4 (Information Efficiency)**.** *For an interaction block Φ_k with computational cost $C_k > 0$, define its information efficiency as*

$$\eta_k := \frac{\Delta_k}{C_k}. \tag{22}$$

This provides a cost-aware criterion for deciding whether an interaction order is worth retaining.

13 Synthetic Validation Protocol

Before applying the framework to learned representations on real datasets, the estimator should be validated on synthetic tasks where the true relevant interaction order is known by construction.

Remark 8 (Choice of R in synthetic experiments)**.** *Throughout this section, we set $R = \varnothing$ (the empty baseline). This is the canonical choice for synthetic validation: there is no learned representation to condition on, and the estimator's task is purely to identify the interaction order from the raw feature blocks Φ_1, Φ_2, Φ_3. The empty baseline avoids the circularity and information-containment issues that arise with learned R, and provides the clearest test of whether the CMI estimator correctly recovers the ground-truth order structure.*

13.1 Ground-Truth Task Families

We recommend four classes of synthetic tasks:

Order-1 task. Let $X \in \mathbb{R}^d$ and define

$$Y = \mathbf{1}\{w^\top X + \varepsilon > 0\}.$$

The predictive structure is essentially linear / unary.

Order-2 task. Let

$$Y = \mathbf{1}\{X_1 X_2 + \varepsilon > 0\}.$$

The target depends on a pairwise interaction.

Order-3 task. Let

$$Y = \mathbf{1}\{X_1 X_2 X_3 + \varepsilon > 0\}.$$

The target depends on a triplet interaction beyond pairwise terms.

Mixed-order task. Let

$$Y = \mathbf{1}\{a^\top X + bX_1 X_2 + cX_1 X_2 X_3 + \varepsilon > 0\},$$

for nonzero coefficients a, b, c.

13.2 Validation Goals

The synthetic validation protocol should verify that:

- for order-1 tasks, Δ_1 dominates and $\Delta_k \approx 0$ for $k \geq 2$,
- for order-2 tasks, Δ_2 is the first substantial gain,
- for order-3 tasks, Δ_3 remains positive after conditioning on lower orders,
- for mixed-order tasks, the estimated order profile matches the construction.

13.3 Estimator Diagnostics

The following diagnostics should be reported:

- mean and standard deviation of $\widehat{\Delta}_k^{\mathrm{CF}}$ across random seeds,
- fold-based uncertainty estimates,
- null distributions obtained via permutation of the candidate block Φ_k,
- sensitivity to auxiliary model choice,
- sensitivity to sample size and dimensionality.

14 Research Plan

This section lays out the logical research program implied by the framework.

14.1 Phase I: Stabilize the Estimation Backbone

The first phase focuses on estimator reliability. The goal is to ensure that the cross-fitted plug-in estimator

$$\widehat{\Delta}_k^{\mathrm{CF}}$$

is stable, interpretable, and capable of recovering the correct interaction order on controlled tasks.

Phase I objectives.

- Formalize the estimand $\Delta_k = I(Y; \Phi_k \mid R, \Phi_{<k})$ cleanly.
- Implement a robust cross-fitted estimator based on held-out log-loss reduction.
- Validate the estimator on synthetic tasks with known true order structure, using $R = \varnothing$.
- Quantify bias, variance, and ranking stability across seeds and folds.

Phase I deliverables.

- A short mathematical note (this document).
- A reusable estimator implementation.
- A synthetic benchmark suite.
- A calibration report showing when the estimator can and cannot be trusted.

14.2 Phase II: Order-2 Representation Analysis

The second phase uses the stabilized estimator to analyze structured order-2 interactions.

Phase II objectives.

- Compare unary vs pairwise representations.
- Decompose order-2 contributions into branch-wise components, e.g. symmetric vs antisymmetric.
- Determine whether order-2 structure contributes unique predictive information beyond the baseline representation.

Remark 9 (Choice of R in Phase II)**.** *Phase II introduces a non-trivial R for the first time. The recommended approach is to use a* frozen unary encoder *as R: a model trained using only Φ_1 (linear / unary features), so that $\Delta_2 = I(Y; \Phi_2 \mid R, \Phi_1)$ cleanly measures the additional value of pairwise structure beyond what a strong unary model already captures. If the analysis is layer-specific (e.g. evaluating at different depths of a neural network), R should be set to the output of the layer immediately preceding the interaction block, and held fixed across all comparisons within Phase II. The key invariant to maintain: R must not have been trained using Φ_2 as part of its input or supervision signal.*

Phase II deliverables.

- An empirical comparison of order-1 and order-2 profiles.
- Branch-wise CMI decomposition.
- A case study on Clifford-style pairwise interactions.

14.3 Phase III: Higher-Order Generalization

The third phase extends the architecture and the analysis to higher-order interaction blocks.

Phase III objectives.

- Define practical feature constructions for $\Phi_3, \Phi_4, \ldots$.
- Estimate whether higher-order interactions add unique information beyond lower orders.
- Determine whether higher-order signal is statistically significant and computationally worthwhile.

Phase III deliverables.

- A hierarchy of higher-order feature blocks.
- Empirical estimates of $\Delta_3, \Delta_4, \ldots$.
- A comparison of cumulative information gain vs computational cost.

14.4 Phase IV: Adaptive Order Selection

The fourth phase uses the order-wise CMI quantities to guide architecture design.

Phase IV objectives.

- Build order-gated models with order-specific branches.
- Use $\widehat{\Delta}_k$ and η_k to determine which orders should be retained.
- Study whether the interaction-order profile can be learned dynamically across tasks.

Phase IV deliverables.

- An adaptive-order interaction architecture.
- A cost-aware order-selection rule.
- A demonstration that the CMI framework can guide architectural pruning or expansion.

15 Decision Roadmap: What Should Be Done Next?

The framework naturally suggests several possible next steps. This section organizes them by logical dependency.

15.1 Option A: Improve the Estimator First

This option prioritizes the statistical backbone.

Why choose this first? Because every later claim — about order-2 usefulness, higher-order relevance, or adaptive-order selection — depends on the reliability of $\widehat{\Delta}_k$.

Recommended when:

- the main uncertainty is whether the current CMI estimation is stable enough,
- synthetic validation has not yet been completed,
- one wants a rigorous measurement tool before committing to architectural development.

Output: A trusted CMI estimation backbone.

15.2 Option B: Analyze Order-2 Interactions First

This option uses the current framework to study pairwise interactions in depth before moving to higher order.

Why choose this first? Because order-2 is the simplest nontrivial case and already includes rich decompositions (e.g. symmetric vs antisymmetric, inner vs wedge).

Recommended when:

- the estimator is already reasonably stable,
- one wants the quickest path to an interpretable empirical result,
- the goal is to connect the framework to Clifford-style interactions directly.

Output: A strong order-2 case study and possibly an early paper.

15.3 Option C: Prototype Higher-Order Blocks Early

This option prioritizes architectural exploration.

Why choose this first? Because higher-order interaction design may surface implementation bottlenecks, computational constraints, or representational ideas that also inform the estimator.

Recommended when:

- the research goal is architecture-first,
- one expects representation design to be the main challenge,
- one is comfortable iterating with provisional estimators.

Output: A family of higher-order interaction models, though with weaker measurement guarantees early on.

15.4 Recommended Logical Next Step

The most logical next step is:

> **Phase I: improve and validate the CMI estimator on synthetic tasks before moving to higher-order architectural claims.**

The reason is simple:

- if the estimator is unstable, all downstream conclusions become fragile;
- if the estimator works well on known-order synthetic tasks, then later empirical claims about order-2, order-3, and adaptive-order selection become much more credible.

Thus the immediate next milestone should be:

> **Immediate milestone.** Implement and benchmark the cross-fitted estimator $\widehat{\Delta}_k^{\mathrm{CF}}$ on synthetic tasks with known ground-truth interaction order (using $R = \varnothing$), and evaluate its stability across auxiliary model classes, folds, seeds, and sample sizes.

16 Practical Milestone Checklist

A concrete sequence for the next iteration is:

1. Finalize this mathematical extension.
2. Implement the order-k estimator in reusable code.
3. Build synthetic datasets with known order structure.
4. Benchmark the estimator on these datasets (with $R = \varnothing$).
5. Decide whether the estimator is reliable enough to support order-2 and higher-order empirical analysis.
6. Only then, proceed to Clifford-style order-2 analysis and higher-order generalization, introducing non-trivial R progressively and documenting its choice explicitly at each stage.